

Pre-work · Appendicular Skeleton

Appendicular Skeleton, the upper and lower limbs. Read both Part A chapters first. Learn the markings vocabulary before the markings, because it repeats on every bone.

Module 2 · 3 of 4

Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

- 1 Start with the reading.**
Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
- 2 Organize what you read.**
In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
- 3 Draw it.**
In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
- 4 Apply it.**
Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
- 5 Come back to it later in the week**
, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

2 min

PANEL 1 OF 3

Turn the two limb chapters into mini visuals. Eight chunks, and the two comparison splits are worth the most.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 GRID

Bone markings vocabulary

Marking down the side, type and example across the top. Nine rows. Sort each as a projection, a depression, or an opening. Do this one first.

2 CHAIN

Upper limb, proximal to distal

Girdle, arm, forearm, hand, with the bones under each link. Then run a second chain underneath for the lower limb and line the two up so the parallels show.

3 HUB

The scapula

Scapula at the hub. Spoke to the three borders, the spine, acromion, coracoid process, glenoid cavity, and the three fossae.

4 SPLIT

Radius against ulna

One split. Side of the forearm, proximal markings, distal markings on each branch. This is the pair most people lose.

5 HUB

The humerus

Humerus down the middle. Proximal markings on one side, distal on the other. Mark which distal surface meets which forearm bone.

6 HUB

The coxal bone

Acetabulum at the hub, because all three bones meet there. Spoke out to ilium, ischium, and pubis, then to the markings on each.

7 SPLIT

Tibia against fibula

One split. Side, whether it bears weight, and the malleolus each one forms. Three branches, and the weight-bearing branch explains the rest.

8 GRID

Carpals and tarsals

Two grids side by side. Carpals in two rows of four with the mnemonic, tarsals in their arrangement with the three C's around the N and T.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

2 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

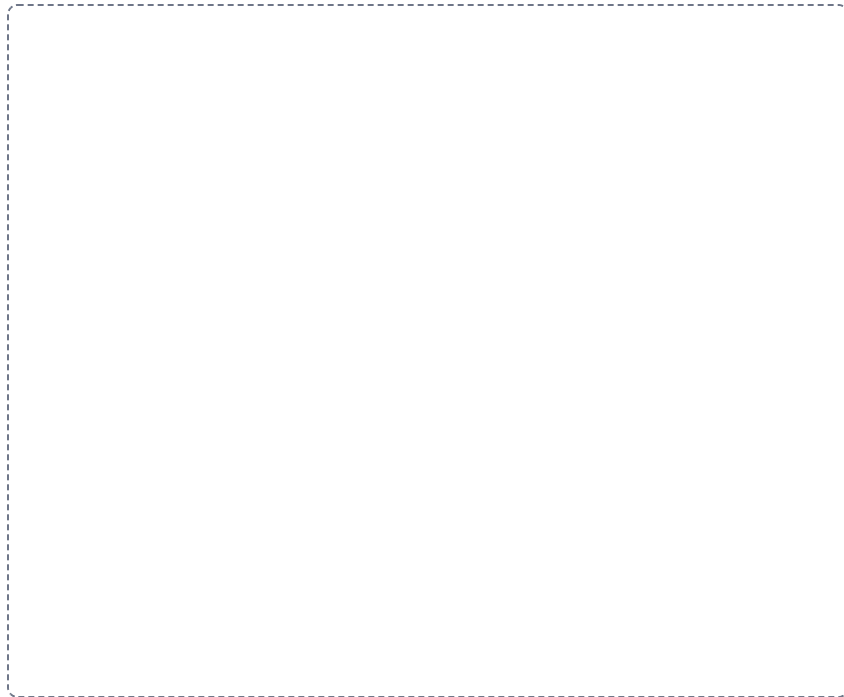
DRAWING 1

The humerus, both views

BRAIN DUMP FIRST

Two minutes. Every marking on the humerus, proximal and distal.

Draw the humerus twice, anterior and posterior. On the anterior view mark head, anatomical and surgical necks, greater and lesser tubercles, intertubercular sulcus, deltoid tuberosity, capitulum, trochlea, and both epicondyles. On the posterior view add the olecranon fossa and the radial groove, and mark where the radial nerve runs.



In your notebook, head the page **M2-7 The humerus, both views** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can say which forearm bone meets the capitulum and which meets the trochlea without pausing.

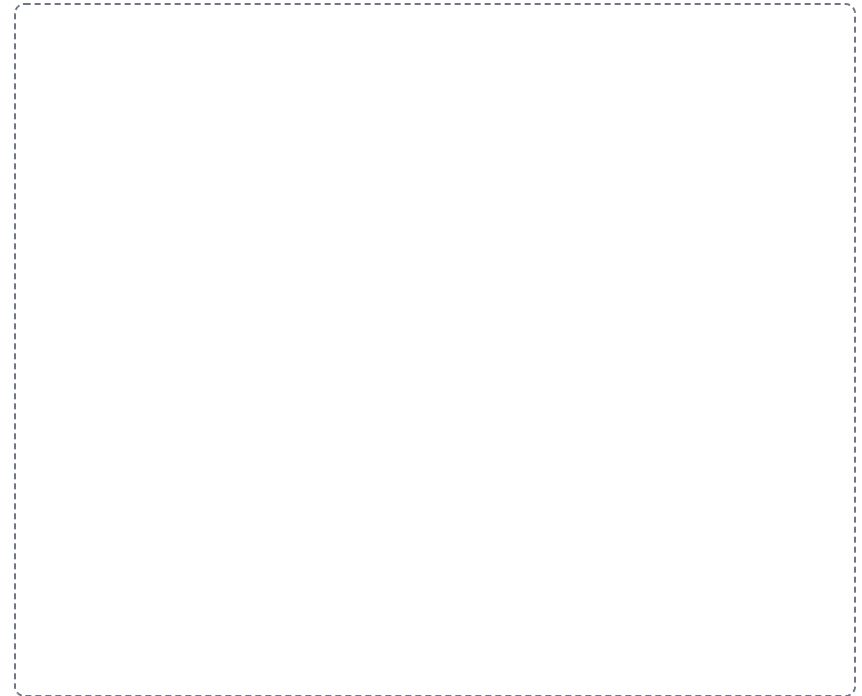
DRAWING 2

The forearm in supination and pronation

BRAIN DUMP FIRST

One minute. Radius and ulna, which is which, and what moves.

Draw the two bones side by side in supination, palm forward, and label radius lateral and ulna medial. Then draw them again in pronation with the radius crossed over the ulna. Mark the proximal and distal radioulnar joints and the interosseous membrane in both.



In your notebook, head the page **M2-8 The forearm in supination and pronation** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain which bone moves and which stays put, and name the joints that allow it.

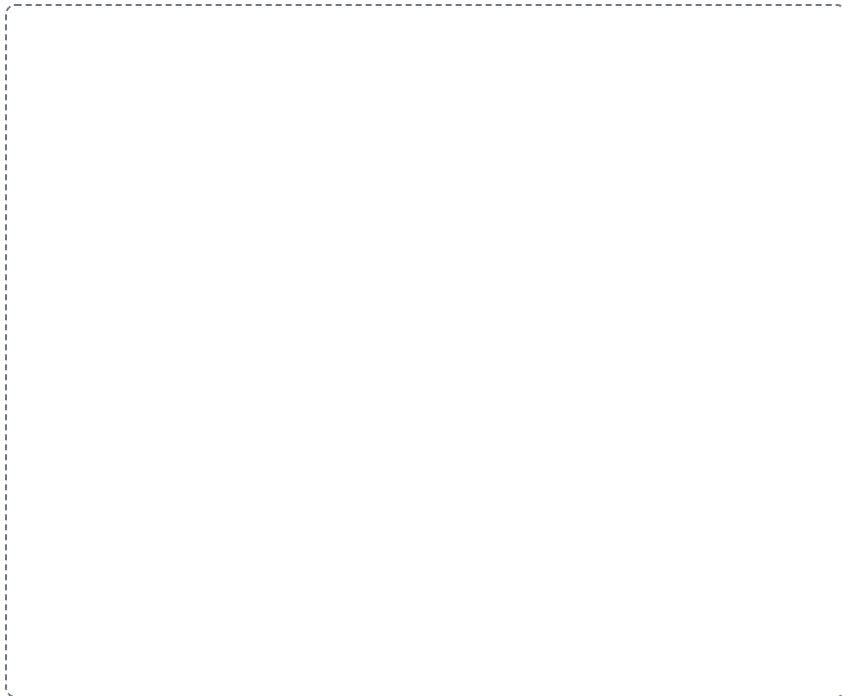
DRAWING 3

The coxal bone and the knee

BRAIN DUMP FIRST

One minute. The three fused bones, then the ligaments of the knee.

Draw the coxal bone from the side, shading ilium, ischium, and pubis differently, and mark the acetabulum where all three meet plus the obturator foramen. Then draw the knee from the front with the patella removed: femoral condyles, tibial plateau, both menisci, both cruciate ligaments crossing, and both collateral ligaments.



In your notebook, head the page **M2-9 The coxal bone and the knee** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can name the three structures of the unhappy triad on your own knee drawing and say what force causes it.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A patient falls on an outstretched hand and has tenderness in the anatomical snuffbox. Name the likely fracture and say why it matters.

Answer. A scaphoid fracture.

Reasoning. Start from the location, then ask what lies under it. The snuffbox is the hollow at the base of the thumb, and the scaphoid sits in its floor, so tenderness there points at that bone. It matters because the scaphoid has a poor blood supply, so a fracture missed on a first X-ray can fail to heal. That is a structural fact producing a clinical rule: suspect it, splint it, image it again.

Set A · Name the marking

1. The only bony attachment of the upper limb to the axial skeleton is the _____.
2. The socket that receives the head of the humerus is the _____.
3. The point of the elbow is the _____ of the _____.
4. The deep cup where the three hip bones meet is the _____.
5. The tarsal that receives the body weight from the tibia is the _____.

Set B · Predict it

1. A fracture at the surgical neck of the humerus. Name the nerve at risk and say why it is checked.

2. A patient cannot feel the little finger after knocking the medial elbow. Name the nerve and where it runs.

3. An older adult falls and cannot bear weight on one leg, with the limb shortened and rotated. Name the likely fracture site and the structure whose blood supply is threatened.

4. Surgeons harvest bone from the fibula rather than the tibia. Explain why, using weight bearing.

Set C · Tell it apart

1. Give one feature that tells the radius from the ulna at the elbow, and one at the wrist.

2. Name the carpals in both rows in order, using the mnemonic, and say which is fractured most often.

3. Compare the pectoral and pelvic girdles by stability and range of motion, and say what causes the difference.

YOU KNOW PANEL 3 WORKED WHEN

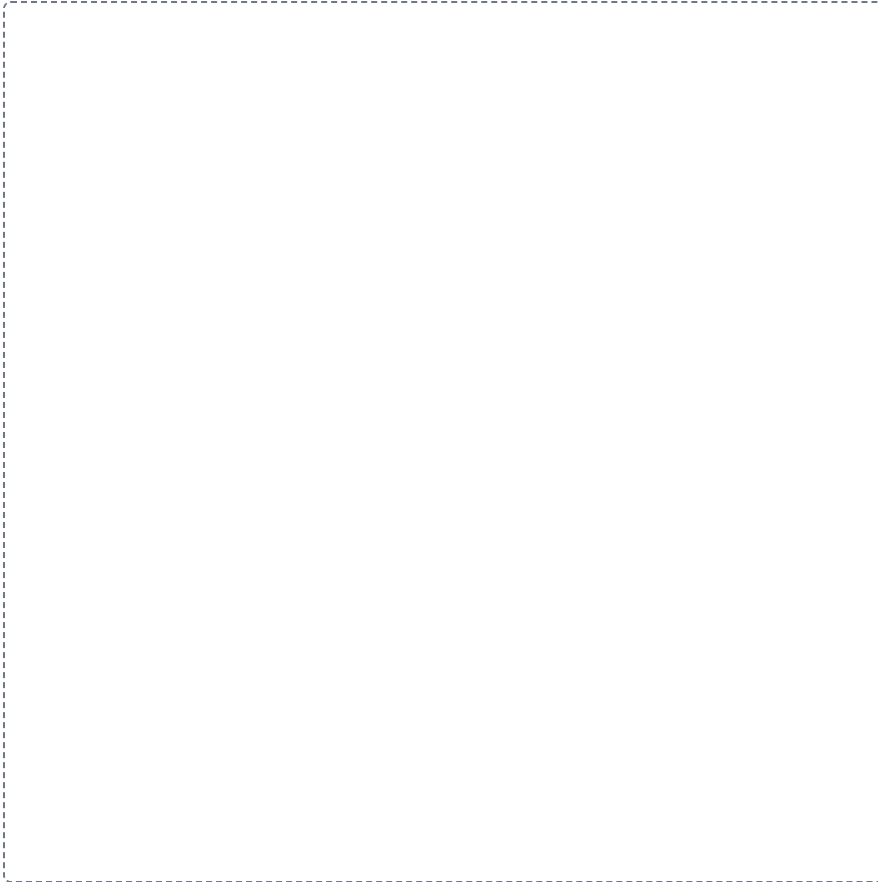
Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

Mini visual sheet**2 min**

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

Bone markings vocabulary**GRID**

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

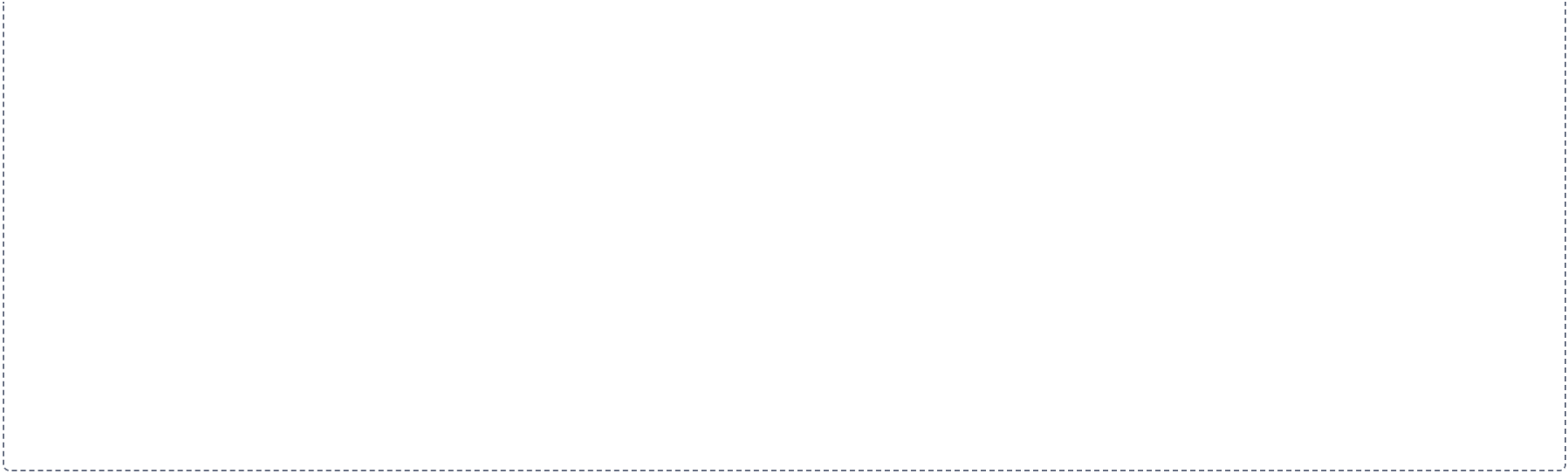


Upper limb, proximal to distal

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

A large dashed rectangular box intended for the student to draw a flowchart showing the steps of the upper limb from proximal to distal. The box is empty and occupies the central portion of the page.

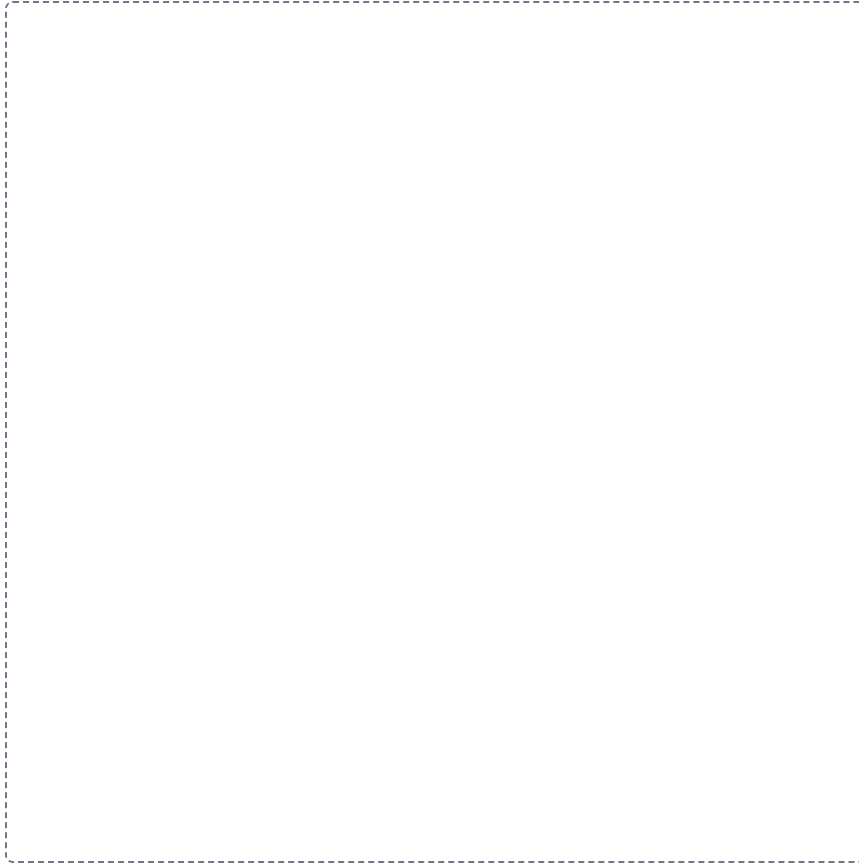


12-3

The scapula

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

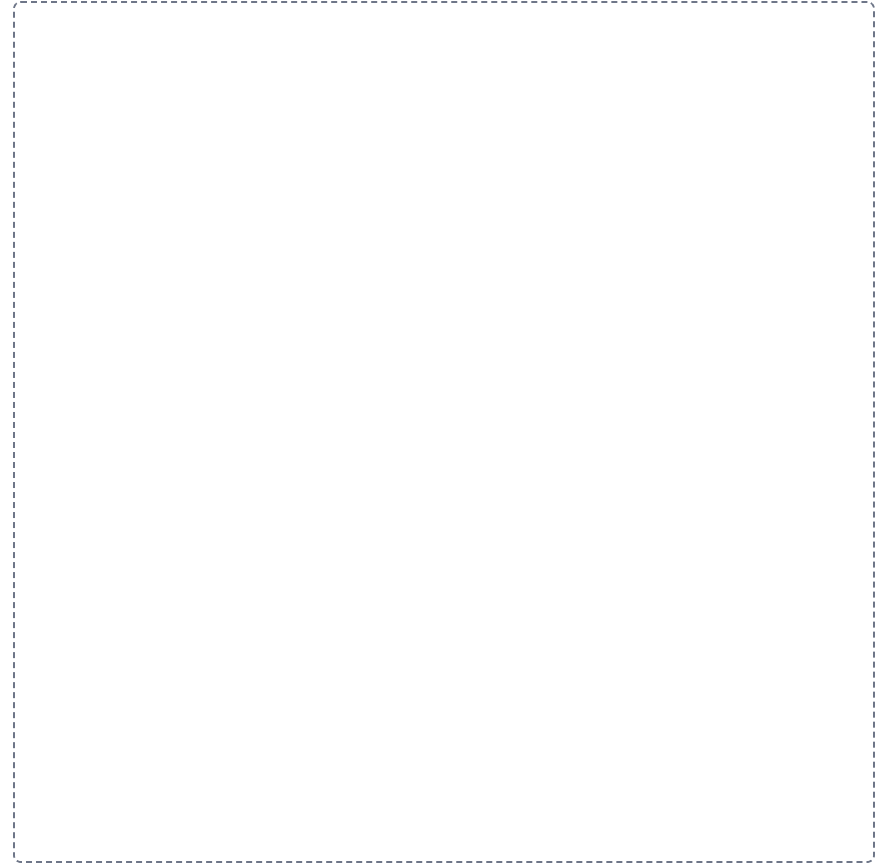


12-4

Radius against ulna

SPLIT

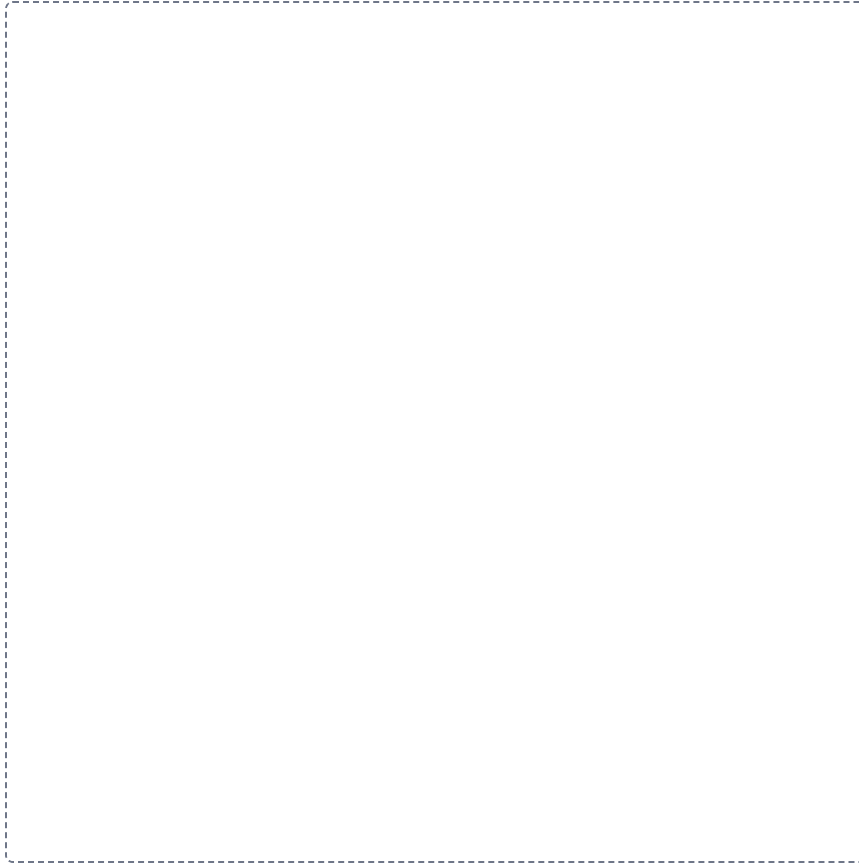
Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.



The humerus

HUB

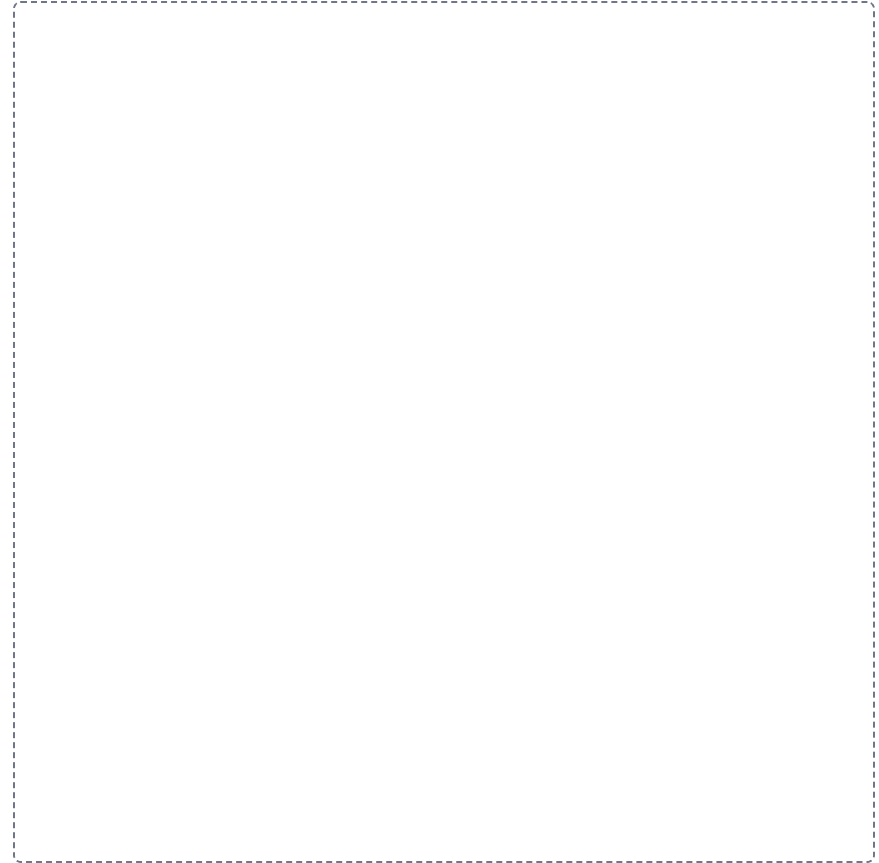
Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.



The coxal bone

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

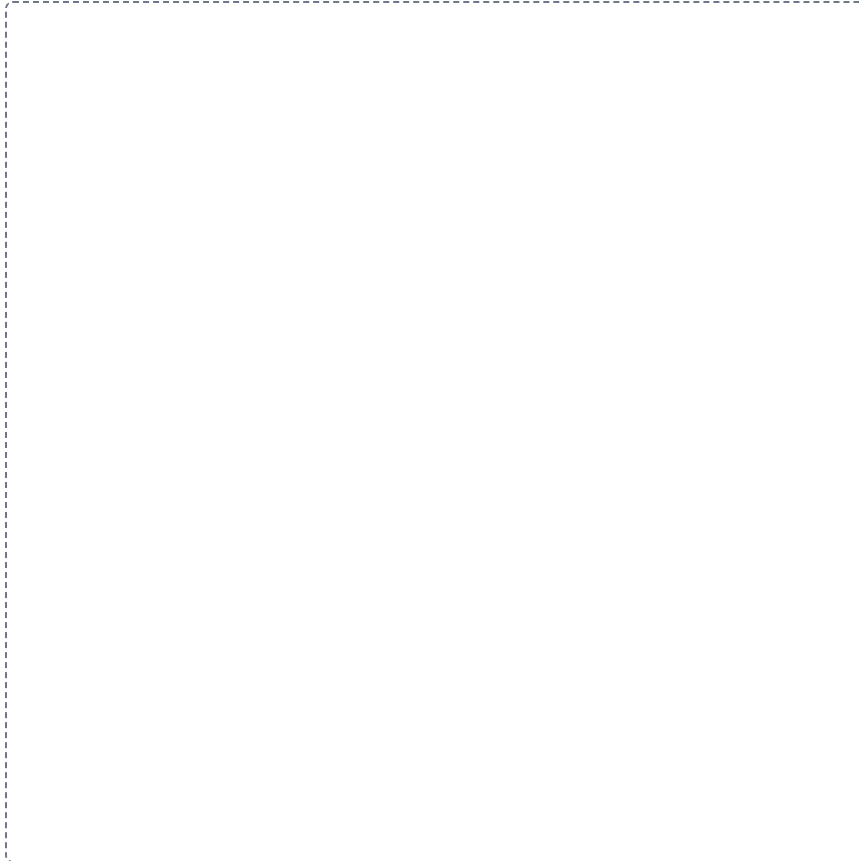


M2-7

Tibia against fibula

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.



M2-8

Carpals and tarsals

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

